JevFusion: Evaluating a Specialized Decision Model for Cost-Efficient Adaptive Hybrid Retrieval

Rihab Atwi
Faculty of Science
Lebanese University
Beirut, Lebanon
rihab112000@gmail.com

Ibrahim Shawraba
Faculty of Science
Lebanese University
Beirut, Lebanon
ibrahim.shawraba11@gmail.com

Sophie Aoun
Faculty of Science
Lebanese University
Beirut, Lebanon
Sophieaoun0@gmail.com

Abstract

DAT (Hsu and Tzeng, 2025) sets a per-query weight between BM25 and dense retrieval by asking a general-purpose LLM to score the top-1 result of each retriever. We reproduce DAT on SQuAD and evaluate JevFusion, which keeps DAT's rubric and weighting rule but replaces the GPT-4o-mini judge with Jev, a specialized decision model. We report three findings. First, Jev judging the top-1 results (J1) raises P@1 over DAT on SQuAD by 1.64 points (95% CI [+1.00, +2.27]) at 2.3 times lower judge cost; on BEIR NFCorpus the difference is inconclusive (+0.53 nDCG@10, CI [−0.28, +1.35]) at 2.6 times lower cost. Second, judging the top-3 results lowers P@1 by 1.90 points on SQuAD and makes no significant difference on NFCorpus. Third, on NFCorpus no judge beats a fixed weight tuned on in-domain dev data.

1 Introduction

Retrieval-augmented generation (RAG) can only use what the retriever finds (Lewis et al., 2020). Sparse retrieval such as BM25 (Robertson and Zaragoza, 2009) matches exact terms; dense retrieval (Karpukhin et al., 2020) matches meaning. Hybrid retrieval combines the two, most simply by a convex combination of normalized scores with one weight α for all queries (Bruch et al., 2023). A fixed weight cannot follow queries for which one retriever is clearly better.

DAT (Hsu and Tzeng, 2025) sets α per query. An LLM scores the top-1 result of each retriever on a 0–5 rubric, and a case rule turns the two scores into α. In the paper, DAT outperforms a fixed weight tuned on validation data, on SQuAD and on the Chinese DRCD, with several judge models including GPT-4o-mini. DAT names its own main limitation: every query needs a call to a general-purpose LLM, which adds latency and cost (Hsu and Tzeng, 2025, Appendix B). It also argues that the top-1 result of each retriever "provides sufficient signal" for the weighting decision.

We test both points. JevFusion keeps DAT's rubric, its α rule and its fusion, and changes the judge to Jev (TypeSafe, 2026), a specialized decision model that answers typed questions with a probability for each level of a scale. **RQ1**: can Jev, judging the same top-1 documents (J1), replace DAT's judge without losing quality, at lower cost and latency? **RQ2**: does judging the top-3 documents of each retriever (J3) give better weights? **RQ3**: are the judges worth their cost compared with zero-call baselines, a tuned fixed α and reciprocal rank fusion (RRF; Cormack et al., 2009)?

We answer them on two datasets with opposite structure. SQuAD (Rajpurkar et al., 2016) has exactly one relevant paragraph per question, and BM25 and dense retrieval are about equally strong. BEIR NFCorpus (Thakur et al., 2021; Boteva et al., 2016) has many graded relevant documents per short medical query, and dense retrieval is much stronger. Our contributions are:

- **A reproduction of DAT on SQuAD**, within 1.2 points of the paper's DAT results on P@1 and MRR@20 and within 2.0 points on α accuracy on the hybrid-sensitive subset. DAT's gain over a tuned fixed weight is significant: +1.64 P@1 points (95% CI [+0.80, +2.47]).
- **A specialized judge.** On SQuAD, J1 improves P@1 over DAT by +1.64 points ([+1.00, +2.27]) at 2.3 times lower judge cost. On NFCorpus the difference is inconclusive (+0.53 nDCG@10, [−0.28, +1.35]) at 2.6 times lower cost. J1 is faster than DAT on 81% and 94% of queries.
- **Top-1 vs. top-3 judging.** Top-3 judging lowers P@1 by 1.90 points on SQuAD and makes no significant difference on NFCorpus. The evidence is consistent with the label structure: with one relevant document, the extra docu-

ments are wrong by construction and dilute the correct retriever's score.

- **Adaptive vs. fixed fusion depends on the dataset.** On NFCorpus no judge beats a fixed α tuned on the dev split, and DAT is significantly below it (-1.32 nDCG@10 points). This is evidence for keeping a tuned global weight unless the judge gives a clear reason to leave it, the design of the gate in Frontier-DAT (pen9rum, 2026).

To our knowledge, as of October 2026, no published study evaluates a specialized decision model as the judge of LLM-scored adaptive fusion, or compares top-1 and top-3 judging under DAT's rule on both a single-relevant and a multi-relevant benchmark.

2 Related Work

Hybrid fusion. Bruch et al. (2023) find that a convex combination of lexical and semantic scores outperforms reciprocal rank fusion (Cormack et al., 2009) and needs only a few training examples to tune its single parameter. Both use one setting for every query. We use the convex combination with min-max normalization, as DAT does, and RRF as a zero-call baseline.

Query-adaptive weights without an LLM. Entropy-based dynamic hybrid retrieval reweights sparse and dense retrieval over several rounds, using normalized Shannon entropy as a confidence proxy (Perez et al., 2025). UMG-RAG uses the entropy of each score list as a training-free reliability estimate for long-document RAG (Jung and Wang, 2026). Query Adaptive Rank Fusion derives a closed-form, training-free weight from IDF statistics, and reports that RRF fails to beat its stronger component on four BEIR corpora, including NFCorpus (Rafeeq et al., 2026). Posokhov et al. (2026) learn to predict α, explicitly to avoid LLM-based online evaluation. In financial document retrieval, Wu and Lin (2026) find oracle headroom for per-query fusion but no reliable gain of three non-LLM routers over a fixed blend. Our NFCorpus result points the same way for LLM-based judges.

LLM-judged adaptive retrieval. DAT (Hsu and Tzeng, 2025) is our baseline (§3). QuDAR (Kim et al., 2026) combines a margin-based confidence (the gap between the top-1 and top-2 scores) with blind LLM relevance scoring, and adapts both between sparse and dense retrieval and between the original and an expanded query. Frontier-DAT (pen9rum, 2026), a public code repository without published results, judges every document that is ranked first for some α and uses a gate to decide whether to leave a tuned global α. MoR (Kalra et al., 2025) mixes sparse, dense and human retrievers with zero-shot weights. RetrievalRouter (Kuru et al., 2026) and Adaptive-RAG (Jeong et al., 2024) route each query to a retrieval pipeline or strategy instead of setting a fusion weight. LLMs are also widely used as judges of relevance and quality (Gu et al., 2024). A different way to spend per-query compute is a cross-encoder reranker, which scores each query–document pair jointly, as in BERT-based passage re-ranking (Nogueira and Cho, 2019); a deployment could rerank the fused list instead of, or after, judging the fusion weight.

How our work differs. We propose no new weighting rule. We hold DAT's rubric, α rule and fusion fixed, change only the judge and the number of documents judged, and measure the effect on two datasets with opposite label structure, against the zero-call baselines a deployment would use.

3 Background: DAT

For a query q, BM25 and dense retrieval each score the candidate documents. DAT min-max normalizes each score list per query, giving $\tilde{S}_{\text{BM25}}(q,d)$ and $\tilde{S}_{\text{dense}}(q,d)$ in $[0,1]$, and ranks documents by

$$R(q,d) = \alpha(q)\,\tilde{S}_{\text{dense}}(q,d) + \big(1-\alpha(q)\big)\,\tilde{S}_{\text{BM25}}(q,d). \tag{1}$$

A fixed-weight hybrid uses one α for all queries, tuned on held-out data.

DAT takes the top-1 document of each retriever and asks an LLM to score both on a rubric from 0 to 5: 5 is a direct hit, 3–4 a "good wrong result" close to the answer, 1–2 a "bad wrong result" that is loosely related but misleading, and 0 is off-track (Hsu and Tzeng, 2025, Appendix A; our copy of the prompt is in Appendix A). With the dense score S_D and the BM25 score S_B, DAT's Eq. 6 is

$$\alpha(q) = \begin{cases} 0.5 & S_D = 0 \text{ and } S_B = 0, \\ 1 & S_D = 5 \text{ and } S_B \neq 5, \\ 0 & S_B = 5 \text{ and } S_D \neq 5, \\ \dfrac{S_D}{S_D + S_B} & \text{otherwise,} \end{cases} \tag{2}$$

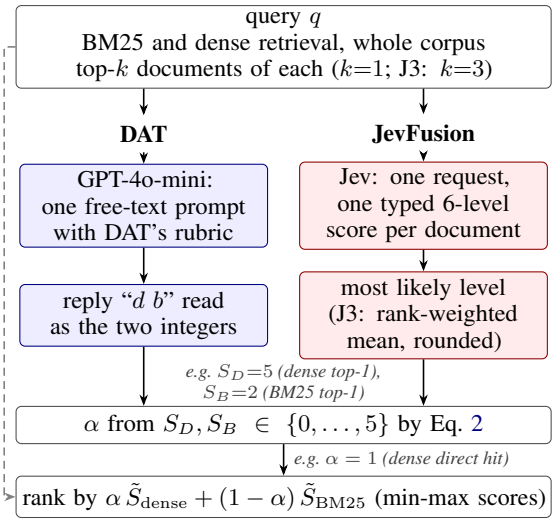


Figure 1: DAT (left lane) and JevFusion (right lane). White boxes are shared; only the shaded boxes differ: the judge, and how its answer becomes the two integer scores S_D and S_B. Dashed: the retrieval scores feed the fusion directly. Example on the arrows: if the dense top-1 document is scored 5 and the BM25 top-1 document 2, Eq. 2 gives $\alpha = 1$.

rounded to one decimal. Equal scores give $\alpha = 0.5$, including two direct hits.

Evaluation in DAT. DAT reports Precision@1 (P@1) and MRR@20 on SQuAD, and two diagnostics that we also use. A question is *hybrid-sensitive* if the rank of its gold paragraph changes somewhere on the grid $\alpha \in \{0, 0.1, \ldots, 1\}$; only these questions can be affected by the choice of α. The α *accuracy* of a method is the share of questions for which its α reaches the best gold rank on that grid. DAT's prose definitions are loose; we use the ones above, which are consistent with the paper's tables (the questions outside the subset contribute identically to every method).

4 JevFusion

JevFusion replaces DAT's judge and leaves everything else unchanged (Figure 1).

Judge. Jev (`jev-1.13.0`) is a decision model served through TypeSafe's System One API (TypeSafe, 2026). A request contains a *state* and typed *questions*. We send one request per query. The state is the query and the judged documents, labelled `dense_1`, `bm25_1` and so on. Each document gets one question of type `score` whose six levels are DAT's rubric, from 0 ("completely off-track") to 5 ("direct hit"); Appendix A gives the exact text. For each document Jev returns a decimal score, a probability for each level and a confidence. A document returned by both retrievers is sent under both labels, as DAT's prompt shows it twice. Replies are validated (model version, one answer per document, probabilities for all six levels); invalid replies are recorded, never replaced by a default.

J1: top-1 judging. J1 judges the same two documents as DAT. A document's integer score is its most likely level (ties go to the lower level), and α follows from Eq. 2. Rounding the decimal score was the alternative; the choice was made on the tuning split. J1 thus differs from DAT in three ways at once: the model, the interface (typed questions with level probabilities instead of a free-text reply), and how the integer is obtained. We compare "DAT's judge" with "Jev using DAT's rubric", the practical replacement question, not a model-against-model test.

J3: top-3 judging. J3 judges the top-3 documents of each retriever in one request. For retriever r with most likely levels $\ell_{r,1}, \ell_{r,2}, \ell_{r,3}$ in rank order, the score is the rank-weighted mean

$$S_r = \text{round}\left(\frac{\ell_{r,1} + \ell_{r,2}/2 + \ell_{r,3}/3}{1 + 1/2 + 1/3}\right), \quad (3)$$

rounded half up; α again follows from Eq. 2. This was the best of three aggregations on the tuning split (Appendix B). With one document per retriever, Eq. 3 reduces to J1, so any difference between J3 and J1 comes from judging documents 2 and 3.

Cost. Jev reads more input tokens than DAT's prompt (on SQuAD, a mean of 1,118 per J1 request against about 724), but costs $0.042 per million input tokens, with output free, against $0.15 for GPT-4o-mini (TypeSafe, 2026; OpenRouter, 2026). Reported costs are for the full test runs: the charge reported by OpenRouter for DAT, and input tokens times the list price for Jev.

5 Experimental Setup

SQuAD. DAT samples whole SQuAD v1.1 articles at random until just under 3,000 questions; the article list and seed were not published. We apply the same rule (seed 42) to the v1.1 dev split: 13 articles, 621 paragraphs and 2,996 test questions (paper: 13, 585 and 2,976). A disjoint tuning set of 13 train-split articles has 623 paragraphs and 2,865 questions. Every question has exactly one gold paragraph.

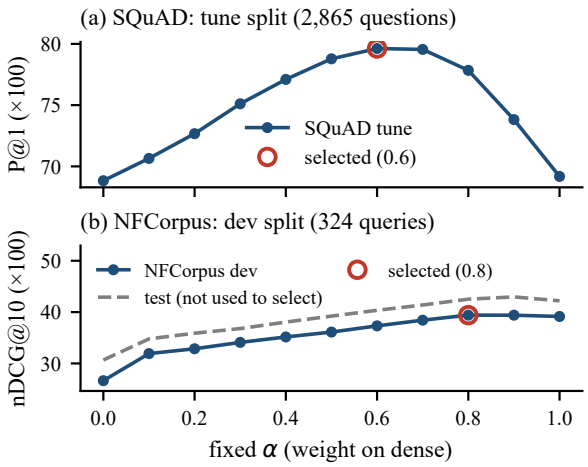


Figure 2: Fixed-α fusion on the tuning splits. (a) SQuAD peaks at 0.6, near DAT's tie value 0.5. (b) NFCorpus peaks at 0.8–0.9; dashed: the test curve, shown for reference and not used for selection.

NFCorpus. The BEIR release (Thakur et al., 2021; Boteva et al., 2016) has 3,633 medical documents (title and abstract), 324 dev and 323 test queries. A test query has on average 38.19 relevant documents (median 16), graded 1 or 2; the median query has two words. The labels come from citation links and are incomplete.

Retrieval. BM25 is `rank_bm25` BM25Okapi (k_1=1.5, b=0.75) over lowercase word tokens, without stemming or stop-word removal. Dense retrieval uses OpenAI's `text-embedding-3-large` (OpenAI, 2024b), as in DAT, with cosine similarity. Both score the whole corpus, so min-max normalization runs over all documents; ties are broken by corpus order. On NFCorpus our unstemmed BM25 reaches 30.67 nDCG@10, against 32.18 for the Anserini regression on the same index (Yang et al., 2017; Anserini, 2026); adding Porter stemming gives 32.18 exactly. We keep the unstemmed version on both datasets.

Judges. DAT uses GPT-4o-mini (OpenAI, 2024a) through OpenRouter, pinned to OpenAI's endpoint, with DAT's prompt, temperature 0, seed 42 and at most 10 output tokens (DAT states no generation settings). The calls were made in October 2026, so the model snapshot may differ from the authors'. Jev is pinned to `jev-1.13.0`. Every reply is cached by its exact request.

Baselines. BM25 only, dense only, RRF (k=60), and fixed-α fusion. On SQuAD the grid search on the tuning set selects α=0.6, the value DAT reports. On NFCorpus we report 0.6 transferred from SQuAD, and 0.8, selected on the dev split by a rule fixed in advance (best dev nDCG@10; Figure 2).

Protocol. Every choice was made on tuning data before any test result was seen: the J1 integer mapping and the J3 aggregation, each by a rule declared in advance (highest tuning P@1, then MRR@20, then the option closest to DAT). J3 was run on the test set as planned although tuning suggested it would lose to J1. On NFCorpus all judges run zero-shot with the settings chosen on SQuAD.

Metrics. On SQuAD: P@1, MRR@20, α accuracy, and P@1 on the 1,056 hybrid-sensitive test questions. On NFCorpus: nDCG@10 (primary, linear gains, identical per query to `pytrec_eval`; Van Gysel and de Rijke, 2018), Recall@10, MRR@10 and P@1, with grade ≥ 1 relevant. All metrics are $\times 100$.

Statistics. All comparisons are paired on the same queries: the mean difference with a 95% percentile bootstrap confidence interval over queries, 10,000 resamples (Efron and Tibshirani, 1993; Koehn, 2004). A difference is significant when its interval excludes zero. We applied no correction for multiple comparisons: each interval is a separate 95% interval, so some of the many comparisons may exclude zero by chance. For SQuAD RQ1 we fixed a margin in advance: J1 *matches* DAT if the lower bound of the P@1 interval is above -1 point, and *beats* it if the bound is above 0. No margin was fixed for NFCorpus, so we report it descriptively. With 323 queries, the smallest nDCG@10 difference detected with 80% power at the 5% level is $2.8\,\mathrm{SD}/\sqrt{323}$, with SD the standard deviation of the per-query differences: 0.96 (J1 vs. J3), 1.18 (J1 vs. DAT) and 1.22 points (J3 vs. DAT); Appendix Table 6 gives it for every comparison.

Latency. Per dataset, all judges answer the same 100 random test queries, one request at a time, with the judge order rotating per query and two warm-up calls per judge discarded. Calls bypass the cache and are timed on the client, network included, for the judge call alone. The datasets were measured on different days, so we compare judges only within a run.

	P@1		MRR@20		α acc.	
Method	paper	ours	paper	ours	paper	ours
All questions (ours n=2,996; paper n=2,976)						
BM25	75.94	76.10	82.23	82.89	79.81	78.94
Dense	73.96	77.30	81.19	84.89	77.89	81.94
Fixed α=0.6	84.61	85.71	89.97	90.87	89.75	89.75
DAT	86.76	87.35	90.93	91.67	91.94	91.69
Hybrid-sensitive subset (ours n=1,056; paper n=1,111)						
BM25	39.06	34.94	54.20	52.88	45.90	40.25
Dense	33.75	38.35	51.43	58.55	40.77	48.77
Fixed α=0.6	62.29	62.22	74.93	75.54	72.55	70.93
DAT	68.05	66.86	77.50	77.79	78.40	76.42

Table 1: DAT reproduction on SQuAD (test), paper (Tables 2–5 of Hsu and Tzeng, 2025) vs. ours. DAT uses GPT-4o-mini. All values $\times$100. Our article sample differs from the authors' (no seed was published).

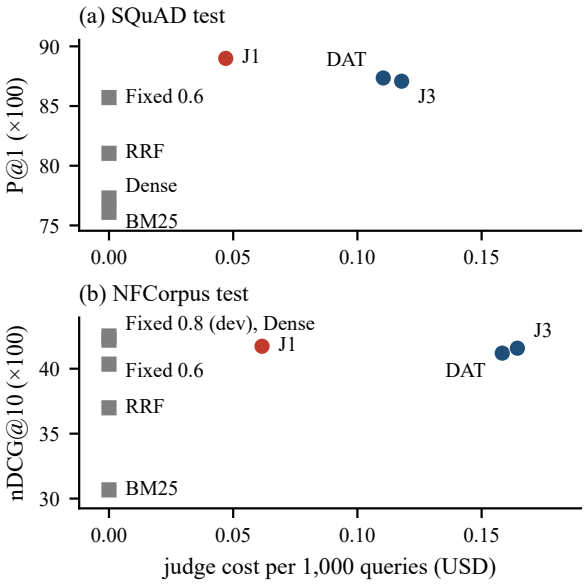


Figure 3: Test quality against judge cost per 1,000 queries; methods without a judge cost nothing. J1 is the cheapest judge on both datasets; only on SQuAD does any judge beat the best fixed weight.

6 Results

6.1 Reproduction of DAT

Table 1 compares our reproduction with the paper. Our DAT reaches 87.35 P@1 and 91.67 MRR@20 (paper: 86.76 and 90.93). On the hybrid-sensitive subset our P@1 is 1.19 points and our α accuracy 1.98 points below the paper's. The largest difference is the dense retriever, 3.34 P@1 points stronger in our setup (77.30 vs. 73.96), for unknown reasons (sample or text formatting). It raises the fixed baseline (85.71 vs. 84.61) and leaves less room for per-query weights: DAT's gain over fixed α=0.6 is smaller than in the paper (+1.64 vs. +2.15 P@1 points) but significant (95% CI [+0.80, +2.47]). As our article sample differs from the authors', we call this a DAT-style reproduction.

6.2 SQuAD

RQ1: J1 reaches 88.99 P@1 (Table 2), +1.64 points over DAT ([+1.00, +2.27]), and +1.01 MRR@20 points ([+0.63, +1.42]). The lower bound is above 0, so under the margin fixed in advance J1 beats DAT. J1 gains rank 1 on 73 questions on which DAT missed it and loses it on 24. **RQ2:** J3 is 1.90 points below J1 ([−2.60, −1.23]) and not distinguishable from DAT (−0.27, [−0.97, +0.47]). **RQ3:** all three judges beat the tuned fixed weight, J1 by +3.27 points ([+2.40, +4.14]). RRF is the weakest fusion, 4.67 points below fixed α.

6.3 NFCorpus

The regime differs from SQuAD (Table 3). Dense retrieval is 11.54 nDCG@10 points ahead of BM25 ([+9.18, +13.95]), and fusion adds little: the dev-tuned α=0.8 is +0.32 points above dense alone ([−0.57, +1.25], not significant), and the SQuAD weight 0.6 is significantly below it (−1.85, [−3.17, −0.54]).

RQ1: J1 is +0.53 points above DAT ([−0.28, +1.35]). The interval contains zero and the difference is below the 1.18 points this comparison can detect, so the result is inconclusive: it shows neither that J1 is better nor that the two are equal. **RQ2:** J3 is 0.15 points below J1 ([−0.82, +0.50]); the SQuAD penalty does not appear, and neither does a gain. **RQ3:** no judge beats the dev-tuned fixed weight. DAT (−1.32, [−2.14, −0.56]) and J3 (−0.94, [−1.70, −0.20]) are significantly below it; J1 is 0.79 points below, with an interval ending at zero ([−1.59, +0.00]). J1 does beat the transferred α=0.6 (+1.38, [+0.42, +2.39]) and RRF (+4.75, [+3.18, +6.38]). No judge differs significantly from dense alone (Appendix Table 6).

6.4 Cost and latency

J1 is the cheapest judge on both datasets (Figure 3): \$0.0470 per 1,000 SQuAD questions against \$0.1104 for DAT (2.3 times less), and \$0.0616 per 1,000 NFCorpus queries against \$0.1583 (2.6 times less). J3 costs 2.5 (SQuAD) and 2.7 (NFCorpus) times as much as J1, in proportion to its extra input tokens. Jev is also faster. On SQuAD the median latency is 2.35 s for J1 and 2.92 s for DAT; paired per question, J1 is faster on 81% of questions, by a median of 527 ms (95% CI [413, 633]).

Method	P@1	MRR@20	Hybrid P@1	ΔP@1 [95% CI] vs. DAT	ΔP@1 [95% CI] vs. fixed 0.6	Judge $/1k q.	Latency p50 (s)
BM25	76.10	82.89	34.94	−11.25 [−12.48, −9.98]	−9.61 [−10.85, −8.41]	0	—
Dense	77.30	84.89	38.35	−10.05 [−11.32, −8.78]	−8.41 [−9.81, −7.04]	0	—
RRF (k=60)	81.04	87.43	48.96	−6.31 [−7.44, −5.17]	−4.67 [−5.64, −3.70]	0	—
Fixed α=0.6	85.71	90.87	62.22	−1.64 [−2.47, −0.80]	—	0	—
DAT	87.35	91.67	66.86	—	+1.64 [+0.80, +2.47]	0.1104	2.92
JevFusion J1	88.99	92.68	71.50	+1.64 [+1.00, +2.27]	+3.27 [+2.40, +4.14]	0.0470	2.35
JevFusion J3	87.08	91.98	66.10	−0.27 [−0.97, +0.47]	+1.37 [+0.60, +2.17]	0.1178	2.54

Table 2: SQuAD test set (2,996 questions). DAT uses GPT-4o-mini. Metrics ×100; Δ = row method minus the reference, paired bootstrap over questions (10,000 resamples). Judge cost per 1,000 questions is from the full test run (§4). Latency: median of the judge call alone, measured sequentially on the same 100 questions for all judges (§5); p95 in Table 8.

Method	nDCG @10	R @10	MRR @10	P @1	ΔnDCG@10 [95% CI] vs. DAT	ΔnDCG@10 [95% CI] vs. fixed 0.8	Judge $/1k q.	Latency p50 (s)
BM25	30.67	15.22	50.97	42.72	—	—	0	—
Dense	42.20	21.08	62.53	54.18	—	—	0	—
RRF (k=60)	36.99	17.87	60.17	52.63	—	—	0	—
Fixed α=0.6 (SQuAD)	40.35	20.39	61.96	53.25	—	—	0	—
Fixed α=0.8 (dev)	42.52	21.10	64.24	56.04	—	—	0	—
DAT	41.20	20.42	63.08	54.80	—	−1.32 [−2.14, −0.56]	0.1583	1.43
JevFusion J1	41.73	20.69	63.72	55.42	+0.53 [−0.28, +1.35]	−0.79 [−1.59, +0.00]	0.0616	0.92
JevFusion J3	41.58	20.73	62.90	54.49	+0.38 [−0.46, +1.27]	−0.94 [−1.70, −0.20]	0.1644	0.97

Table 3: BEIR NFCorpus test set (323 queries), zero-shot judges. DAT uses GPT-4o-mini. Metrics ×100; Δ = row method minus the reference, paired bootstrap over queries (10,000 resamples). Fixed α=0.8 was selected on the dev split. Latency as in Table 2, measured in a separate run (compare within a table only).

On NFCorpus the medians are 0.92 s and 1.43 s, and J1 is faster on 94% of queries, by a median of 496 ms ([459, 528]). Asked the same request again, J1 repeats its judgment on 95% (SQuAD) and 89% (NFCorpus) of queries, DAT on 94% and 96% (Appendix Table 8).

7 Analysis

7.1 Why J1 beats DAT on SQuAD

On 2,001 of the 2,996 SQuAD questions both retrievers return the same top-1 paragraph, so no α can change rank 1. J1's whole gain over DAT comes from the 774 questions on which exactly one retriever's top-1 is the gold paragraph. There, J1 reaches 95.31 P@1 when only dense is right and 95.39 when only BM25 is right, against 87.16 and 91.06 for DAT. Both judges lean α toward the right retriever about equally often (J1 88.9%, DAT 88.2%; Table 4, Appendix Table 7), but J1 leans the wrong way far less often (1.3% vs. 8.4%). When unsure, J1 gives both documents the same level, which leaves α at 0.5 (9.8% vs. 3.4%). When sure, it is more decisive: the "direct hit" cases of Eq. 2 ($\alpha \in \{0, 1\}$) fire for 802 questions with J1 and 463 with DAT. A possible mechanism, which we did not test, is that Jev answers with a distribution over the levels and we keep the most likely one, whereas a chat model generates two digits directly.

The pattern does not carry over to NFCorpus. On the 83 queries where exactly one top-1 document is relevant, J1 leans the right way less often than DAT (68.7% vs. 79.5%) and the wrong way about as often (15.7% vs. 14.5%). Its small, non-significant edge comes from giving dense, the stronger retriever, more weight: on the 101 queries where J1 does so relative to DAT, it is better on 45 and worse on 23.

7.2 Why top-3 judging hurts on SQuAD only

On SQuAD, a retriever whose top-1 is the gold paragraph has, by construction, wrong documents at ranks 2 and 3, and Jev scores them low: on the tuning set, Jev's mean levels for ranks 1–3 are then 4.94, 2.21 and 1.64 for dense and 4.95, 1.30 and 0.91 for BM25. Eq. 3 pulls the correct retriever's 5 toward 3, while the other retriever often holds the gold at rank 2 or 3 and is pulled up. On the 774 one-right test questions, J3 leans the right way

	SQuAD (P@1)	NFCorpus (nDCG@10)
Test queries	2,996	323
Relevant per query	1	38.19 (med. 16)
Query length (words)	10.1	3.3 (med. 2)
BM25 / dense	76.10/77.30	30.67/42.20
Same top-1 document	66.8%	26.9%
Best fixed α (tuning)	0.6	0.8
DAT − tuned fixed α	+1.64	−1.32
	[+0.80, +2.47]	[−2.14, −0.56]
J1 − DAT	+1.64	+0.53
	[+1.00, +2.27]	[−0.28, +1.35]
J3 − J1	−1.90	−0.15
	[−2.60, −1.23]	[−0.82, +0.50]
α=0.5 share, %	67.9/70.9/50.8	35.0/46.7/36.8
Wrong way, %	8.4/1.3/9.8	14.5/15.7/14.5

Table 4: The two regimes, on the test sets. Metrics and differences $\times 100$; under each difference, its 95% paired bootstrap CI. The last two rows give DAT/J1/J3. "Wrong way": share of the queries whose two top-1 documents differ and exactly one is relevant (774 SQuAD, 83 NFCorpus) on which α leans toward the other retriever.

on only 62.0% (J1: 88.9%). The aggregation, not Jev, causes the drop: Jev gives the top-1 document the same level in J1 and inside a J3 request for 92% of questions. J3 helps only when both top-1 paragraphs are wrong (221 questions: 12.67 P@1 against 7.69 for J1 and DAT), because it pushes α to the extremes less often.

On NFCorpus, when a retriever's top-1 is relevant, its rank-2 and rank-3 documents are also relevant 60.0% and 49.1% of the time for dense and 55.1% and 39.1% for BM25, and Jev's levels fall much less with rank (4.43, 3.31, 2.99 for dense). The dilution is weaker and the penalty disappears. These observations are consistent with the label structure being the cause; we did not test it by intervention. DAT's claim that the top-1 result gives sufficient signal holds in the sense that top-3 judging never helped; our data do not show that top-1 judging is sufficient for nDCG@10, which scores ten ranks.

7.3 When does a judge beat a fixed weight?

On SQuAD the two retrievers are about equally strong, a tuned fixed blend beats the best single retriever by 8.41 P@1 points, and the best fixed weight (0.6) is close to the 0.5 that Eq. 2 gives for every tie (Table 4, Figure 2a). On NFCorpus dense dominates, the two retrievers share the top-1 document on only 26.9% of queries, and the best weight is 0.8. The judges tie, and so set α=0.5, on 35.0% (DAT) to 46.7% (J1) of NFCorpus queries, each time giving BM25 too much weight. They also rarely move toward BM25 when it would help: on the 60 queries where BM25's own ranking is better, they lean toward it on only 16.7% (J1) to 26.7% (DAT).

Room for per-query weights exists on both datasets. Choosing the best grid α per query with the labels (an upper bound, not a method) would reach 91.62 P@1 on SQuAD, +5.9 points over the fixed weight, and 46.22 nDCG@10 on NFCorpus, +3.7 points over the dev-tuned weight. The judges capture part of this on SQuAD and none on NFCorpus. Even on SQuAD, the fixed weight is best when both top-1 paragraphs are wrong (19.91 P@1 against at most 12.67 for the judges). This is evidence for keeping a tuned global weight and leaving it only when the judge gives a clear reason, as the gate in Frontier-DAT does (pen9rum, 2026); we did not test such a gate.

7.4 Do the judges agree with the labels?

Over the 646 judged top-1 NFCorpus documents, Spearman's correlation between judge score and official grade is 0.433 for DAT and 0.473 for J1's most likely level (0.494 for its decimal score). Over all 1,938 documents J3 judged, including ranks 2 and 3, it is 0.511; this covers a different document set and is not directly comparable. Jev separates highly from partially relevant documents (mean level 4.55 for grade 2 vs. 4.09 for grade 1); DAT does not (4.02 vs. 4.09). All judges give 4 or 5 to about a third of grade-0 documents, some of which may be unlabeled relevant ones. Under Eq. 2, better agreement with the labels did not yield better rankings.

8 Conclusion

We reproduced DAT on SQuAD and replaced its general-purpose judge with Jev, a specialized decision model, keeping DAT's rubric, weighting rule and fusion. Judging the top-1 documents, Jev is a cheaper and faster replacement for GPT-4o-mini on both datasets: 2.3 and 2.6 times lower judge cost, and faster on 81% and 94% of queries. On SQuAD it is also more accurate (+1.64 P@1 points), mainly because it rarely pushes α toward the wrong retriever. On NFCorpus its difference to DAT is inconclusive. Judging the top-3 documents never helped: it hurt on SQuAD, where only one document is relevant, and made no significant difference on NFCorpus. The advantage of judged adaptive

fusion over fixed fusion depends on the dataset. It holds on SQuAD, but on NFCorpus no judge beats a fixed weight tuned on dev data. A practical system should therefore compare against a tuned fixed weight, and use the judge to decide when to leave it. Testing such a gate, a neutral weight other than 0.5, and the judges on more datasets are the next steps.

Limitations

Scope. We evaluate two English datasets, retrieval quality only, and not the quality of generated answers. We did not run DAT's second dataset (DRCD) or its other judge models, and we did not compare with a cross-encoder reranker over the fused candidates, an alternative a deployment could use. The NFCorpus test set has 323 queries; differences between judges below about one nDCG@10 point (0.96–1.22, depending on the comparison) cannot be detected reliably, so our non-significant NFCorpus results are inconclusive, not evidence of equality.

Reproduction. Our SQuAD article sample differs from the authors' (no seed was published), our dense retriever is stronger than theirs, and the GPT-4o-mini snapshot we called in October 2026 may differ from theirs. Our BM25 is unstemmed, which lowers it on NFCorpus (30.67 vs. 32.18 with stemming).

What J1 changes. J1 changes the model, the interface and the way the integer score is obtained at the same time; we did not separate these effects. Neither judge's prompt or rubric was optimized; both use DAT's rubric.

Labels and design. NFCorpus labels are incomplete, so a judge that finds unlabeled relevant documents may be penalized. The judges were run zero-shot on NFCorpus with settings chosen on SQuAD, while the fixed weight 0.8 used NFCorpus dev data; this asymmetry is by design but favors the fixed weight. The mechanisms we describe in §7 are observational and were not tested by intervention.

Measurement. Latency was measured from one client location on 100 queries per dataset and includes the network. Results are for `jev-1.13.0` and the October 2026 GPT-4o-mini; costs use October 2026 prices. Jev is a commercial model: its internals are not public, and our description of it relies on the vendor's documentation.

Ethics Statement

We use two public research datasets, SQuAD (Wikipedia paragraphs) and BEIR NFCorpus (medical documents); neither contains private user data. A retrieval system built on medical documents should not be used for medical advice without expert oversight. Both judges are commercial models accessed through paid APIs, and the whole project cost $1.7144 in API fees. Our description of Jev is limited to its public documentation.

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A Judge Prompts

DAT. The DAT judge uses the prompt of Hsu and Tzeng (2025, Appendix A), reproduced here with credit to its authors. The PDF prints it in a code font that adds spaces around punctuation; we restored normal spacing and changed no words. The two placeholders are filled with the top-1 dense and BM25 documents.

```
You are an evaluator assessing the retrieval effectiveness
      of dense retrieval (Cosine Distance) and BM25
      retrieval for finding the correct answer.

## Task:
Given a question and two top1 search results (one from
      dense retrieval, one from BM25 retrieval), score each
      retrieval method from **0 to 5** based on whether the
      correct answer is likely to appear in top2, top3, etc.

### **Scoring Criteria:**
1. **Direct hit --> 5 points**
- If the retrieved document directly answers the question,
      assign **5 points**.
2. **Good wrong result (High likelihood correct answer is
      nearby) --> 3-4 points**
- If the top1 result is **conceptually close** to the
      correct answer (e.g., mentions relevant entities,
      related events, partial answer), it indicates the
      search method is in the right direction.
- Give **4** if it's very close, **3** if somewhat close.
3. **Bad wrong result (Low likelihood correct answer is
      nearby) --> 1-2 points**
- If the top1 result is **loosely related but misleading**
      (e.g., shares keywords but changes context), correct
      answers might not be in top2, top3.
- Give **2** if there's a small chance correct answers are
      nearby, **1** if unlikely.
4. **Completely off-track --> 0 points**
- If the result is **totally unrelated**, it means the
      retrieval method is failing.
---
### **Given Data:**
- **Question:** "{question}"
- **dense retrieval Top1 Result:** "{vector_reference}"
- **BM25 retrieval Top1 Result:** "{bm25_reference}"
---
### **Output Format:**
Return two integers separated by a space:
- **First number:** dense retrieval score.
- **Second number:** BM25 retrieval score.
- Example output: 3 4
(Vector: 3, BM25: 4)
**Do not output any other text.**
```

Jev. One request per query. The `state` holds the question and the judged documents; there is one `score` question per document, all with the same six levels. For J1 the slots are `dense_1` and `bm25_1`; for J3, `dense_1`–`dense_3` and `bm25_1`–`bm25_3`.

```
{"model": "jev-1.13.0",
 "state": {"question": "<question>",
           "passages": {"dense_1": "<text>", "bm25_1":
      "<text>"}},
 "questions": {
   "dense_1": {"type": "score",
     "instructions": "Score the retrieval effectiveness of
      passage 'dense_1', the top1 result of dense retrieval,
      for finding the correct answer to the question: is the
      correct answer likely to appear in this result or in
      the results ranked right after it?",
     "criteria": [<the six levels below>]},
   "bm25_1": {... same, for 'bm25_1', BM25 retrieval ...}}}
```

The six levels, from low to high:

```
0 - Completely off-track: the passage is totally unrelated
      to the question; the retrieval method is failing.
1 - Bad wrong result: loosely related but misleading (e.g.
      shares keywords but changes context); the correct
      answer is unlikely to be nearby.
2 - Bad wrong result: loosely related but misleading; there
      is a small chance the correct answer is nearby.
3 - Good wrong result: somewhat conceptually close to the
      correct answer (e.g. mentions relevant entities or
      related events); the search is in the right direction.
4 - Good wrong result: very conceptually close to the
      correct answer, or a partial answer; the correct
      answer is likely nearby.
5 - Direct hit: the passage directly answers the question.
```

B Choices Made on Tuning Data

Table 5 lists every choice made before the test runs. The two J1 mappings are practically tied on the tuning set and the declared rule picked the most likely level. All three J3 options were already below J1 on the tuning set. Option A is the plain mean of levels; option B maps the mean decimal score to $[0, 1]$ and uses a smoothed ratio instead of Eq. 2.

C All Paired Comparisons

Table 6 gives every paired comparison with all metrics. "Rank 1 gained / lost" counts the questions on which the first method has the gold paragraph at rank 1 and the second does not, and the reverse.

D Judge Behavior and Latency

Table 7 gives the direction of α on the queries where the two top-1 documents differ and exactly one is relevant. Table 8 gives the latency measurement.

E Reproducibility

Code and results are available on request. All runs are cached by exact request, so every number in this paper can be regenerated without new API calls; the tables in this paper are generated by a script from the result files. Software: Python 3.13.3 with `numpy`, `rank-bm25` and `requests`. Models: `text-embedding-3-large` and GPT-4o-mini through OpenRouter (pinned to OpenAI's endpoint), and `jev-1.13.0` through TypeSafe's API, all called in October 2026. Seeds: 42 for the SQuAD sample and the DAT judge, 0 for the NFCorpus latency queries; 10,000 bootstrap resamples. The total API cost was $1.7144 (NFCorpus: $0.3310), within a project cap of $20.

α	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
SQuAD tune P@1	68.83	70.65	72.67	75.11	77.10	78.78	79.62	79.55	77.84	73.82	69.18
NFCorpus dev nDCG@10	26.64	31.92	32.86	34.10	35.16	36.11	37.32	38.43	39.41	39.40	39.15

SQuAD tune (2,865 questions)	P@1	MRR@20	α acc.	Hyb. P@1
J1 mapping: `top_level` (chosen)	83.63	88.52	89.91	66.01
J1 mapping: `round`	83.56	88.52	89.60	65.86
Fixed α=0.6	79.62	86.11	84.57	57.26
J1 (reference)	83.63	88.52	89.91	66.01
J3 option A	80.98	87.19	86.91	60.23
J3 option B	81.26	87.33	86.21	60.84
J3 option C (chosen)	81.47	87.57	87.89	61.29

Table 5: Choices fixed on tuning data before any test result was seen (values $\times 100$). Top: fixed-α grid (selected: 0.6 on SQuAD tune, 0.8 on NFCorpus dev). Bottom: Jev J1 score mapping and J3 aggregation option, selected by tune P@1, then MRR@20, then the option closest to DAT.

SQuAD (2,996 questions)	ΔP@1 [95% CI]	ΔMRR@20 [95% CI]	Rank 1 gained / lost
DAT − fixed 0.6	+1.64 [+0.80, +2.47]	+0.79 [+0.29, +1.30]	107 / 58
J1 − DAT	+1.64 [+1.00, +2.27]	+1.01 [+0.63, +1.42]	73 / 24
J1 − fixed 0.6	+3.27 [+2.40, +4.14]	+1.81 [+1.28, +2.35]	139 / 41
J3 − J1	−1.90 [−2.60, −1.23]	−0.70 [−1.10, −0.30]	28 / 85
J3 − DAT	−0.27 [−0.97, +0.47]	+0.31 [−0.12, +0.76]	58 / 66
J3 − fixed 0.6	+1.37 [+0.60, +2.17]	+1.10 [+0.64, +1.57]	91 / 50
Fixed 0.6 − RRF	+4.67 [+3.70, +5.64]	+3.45 [+2.87, +4.03]	184 / 44
DAT − RRF	+6.31 [+5.17, +7.44]	+4.24 [+3.53, +4.95]	252 / 63
J1 − RRF	+7.94 [+6.81, +9.08]	+5.26 [+4.52, +6.01]	284 / 46
J3 − RRF	+6.04 [+4.87, +7.18]	+4.55 [+3.83, +5.27]	252 / 71

NFCorpus (323 queries)	ΔnDCG@10	ΔR@10	ΔMRR@10	ΔP@1	Detectable
J1 − DAT	+0.53 [−0.28, +1.35]	+0.27 [−0.27, +0.83]	+0.65 [−1.03, +2.35]	+0.62 [−2.17, +3.41]	1.18
J1 − J3	+0.15 [−0.50, +0.82]	−0.04 [−0.42, +0.32]	+0.82 [−0.67, +2.37]	+0.93 [−1.55, +3.41]	0.96
J3 − DAT	+0.38 [−0.46, +1.27]	+0.32 [−0.13, +0.78]	−0.18 [−1.57, +1.14]	−0.31 [−2.79, +1.86]	1.22
J1 − fixed 0.6	+1.38 [+0.42, +2.39]	+0.30 [−0.24, +0.86]	+1.76 [−0.33, +3.93]	+2.17 [−1.24, +5.57]	1.41
J1 − fixed 0.8	−0.79 [−1.59, +0.00]	−0.41 [−1.04, +0.19]	−0.52 [−2.32, +1.22]	−0.62 [−3.72, +2.17]	1.14
J1 − RRF	+4.75 [+3.18, +6.38]	+2.82 [+1.78, +3.99]	+3.55 [+0.57, +6.57]	+2.79 [−1.55, +7.12]	2.28
J1 − dense	−0.47 [−1.51, +0.57]	−0.39 [−0.97, +0.14]	+1.20 [−0.88, +3.28]	+1.24 [−1.86, +4.33]	1.48
DAT − fixed 0.8	−1.32 [−2.14, −0.56]	−0.68 [−1.20, −0.21]	−1.16 [−2.64, +0.27]	−1.24 [−3.72, +1.24]	1.11
DAT − dense	−1.00 [−2.18, +0.21]	−0.67 [−1.37, +0.03]	+0.55 [−1.81, +2.88]	+0.62 [−2.79, +4.02]	1.71
J3 − fixed 0.8	−0.94 [−1.70, −0.20]	−0.37 [−0.90, +0.17]	−1.34 [−2.99, +0.27]	−1.55 [−4.33, +1.24]	1.08
J3 − dense	−0.62 [−1.72, +0.48]	−0.35 [−0.94, +0.23]	+0.37 [−1.96, +2.69]	+0.31 [−3.10, +3.72]	1.57

Table 6: All paired comparisons (values $\times 100$, 95% bootstrap CIs, 10,000 resamples). "Detectable": the nDCG@10 difference this NFCorpus sample detects with 80% power at the 5% level ($2.8\,\mathrm{SD}/\sqrt{323}$).

Data	Judge	Right	Neutral	Wrong
SQuAD	DAT	88.2	3.4	8.4
SQuAD	J1	88.9	9.8	1.3
SQuAD	J3	62.0	28.2	9.8
NFCorpus	DAT	79.5	6.0	14.5
NFCorpus	J1	68.7	15.7	15.7
NFCorpus	J3	65.1	20.5	14.5

Table 7: Direction of α (%) on queries where the two top-1 documents differ and exactly one is relevant (SQuAD 774, NFCorpus 83): toward the relevant retriever, neutral (α=0.5), or the wrong way. DAT uses GPT-4o-mini; J1 and J3 use Jev.

Data	Judge	p50	p95	max	OK	Same
SQuAD	DAT	2.92	4.57	10.67	100	94
SQuAD	J1	2.35	4.14	5.64	100	95
SQuAD	J3	2.54	4.33	6.42	100	80
NFCorpus	DAT	1.43	2.57	4.04	100	96
NFCorpus	J1	0.92	1.66	2.86	100	89
NFCorpus	J3	0.97	1.59	2.35	100	78

Table 8: Sequential judge latency in seconds on 100 test queries per dataset (two separate runs on different days), calls that succeeded, and the share (%) of fresh replies identical to the cached reply. DAT uses GPT-4o-mini; J1 and J3 use Jev.